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Do Paper Markets Price Physical Scarcity? Futures, Positioning, and Delivery Across Metals Markets

Kilambi Research

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Abstract. Whether futures, spot, and physical markets price metals scarcity or something else is a claim made routinely and tested rarely. This paper tests it on the paper side: the shapes of futures curves, the composition of speculative and commercial positioning, the mechanics and statistics of delivery, physical premiums by geography, and a decomposition of monthly returns into macro factors versus everything else. On the canonical numbers used here, gold and silver curves are in clear contango (+472 and +479 basis points annualized, CME, late September 2026), which is inconsistent with acute nearby scarcity; copper's COMEX curve is in steeper contango (+603 basis points) even while the LME prompt touched +$65/t backwardation, a regional and policy-driven split, not a global shortage; gold swap dealers are gross short 60.7 percent of open interest but the CFTC categories do not reveal offsets, so that number is not evidence of bearish exposure or manipulation. Monthly USD and real-rate changes explain 18 to 41 percent of return variance across six metals (n = 55, Modelled C), with tin the most macro-led and zinc unidentified; the residual is unexplained, not evidence of physical tightness. The documented 2022–2026 dislocation episodes (LME nickel 2022, COMEX copper May 2024, gold and silver lease spikes, the 2026 LME copper prompt) were liquidity, conversion, or policy events, and each cleared or was resolving within weeks to months. The companion paper establishes the physical balances; this paper's verdict is that paper markets are, on the whole, pricing macro risk and regional plumbing rather than physical depletion.

Keywords: futures markets, commodity pricing, term structure, CFTC positioning, backwardation, lease rates, physical premiums, Kilambi Research Working Paper No. KR-WP-2026-14

JEL Classification: G13, G15

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1 Introduction

Two prices claim to describe the same metal: the futures price on an exchange and the physical price paid for metal that moves. When the two disagree about scarcity, the disagreement is the market's most quoted and least examined signal. Contango is read as ample supply, backwardation as tightness, a lease-rate spike as a smoking gun of vanishing metal, and a large bank short as manipulation. Each of those readings can be right. Each can also be wrong, and the paper market has a documented habit of being read wrong.

This paper is the second half of a two-paper program on metals markets. The companion paper (Kilambi Research, 2026) builds the physical side: mine-by-mine supply, demand disaggregation, annual balances, and inventory cover for the six core metals (copper, silver, gold, zinc, nickel, tin). This paper takes those physical facts as given inputs and asks the paper-side question: do futures curves, positioning, delivery mechanics, physical premiums, and the macro structure of returns price the physical scarcity the companion paper measures, or are they pricing something else?

The question matters now because the 2022–2026 window produced an unusual density of episodes that look like dislocation: the LME nickel short squeeze of March 2022, the COMEX copper squeeze of May 2024, gold and silver lease-rate spikes in early and late 2025, a Section 232 tariff investigation that pulled record copper stocks into the United States, and an LME copper prompt backwardation in September 2026. Whether those episodes reveal physical depletion or merely paper plumbing decides how one reads every curve, every premium, and every positioning number.

Three findings anchor this paper. First, the curves are not pricing nearby scarcity. Gold and silver futures are in clear contango on official CME settlements as of late September 2026 (+472 and +479 basis points annualized, (Observed, A)); copper's COMEX curve is in steeper contango (+603 basis points, (Observed, B)) at the same time the LME prompt flipped to backwardation. A market pricing acute physical shortage inverts its curve; this one has not, and the copper split is regional and policy-driven, not global. Second, the most cited positioning statistic in metals commentary, the gross short of swap dealers in gold (250,752 contracts, 60.7 percent of open interest on September 22, 2026, (Observed, A)), is not evidence of directional bearish exposure or manipulation: dealers intermediate producer hedges and investor longs, and the CFTC categories do not reveal the offsets, so the honest statement stops at "dealers are net short on the exchange." Third, monthly changes in the US dollar and the 10-year TIPS real yield explain 18 to 41 percent of monthly return variance across the six metals (n = 55, (Modelled, C)), with tin the most macro-led ($R2 = 0.405$) and zinc unidentified ($R2 = 0.194$); the unexplained remainder is reported as unexplained, not relabeled as physical tightness.

The paper makes no claim about any single trader's intent and advances no manipulation thesis. The scope excludes mine-by-mine supply, demand disaggregation, balance construction, inventories in tonnes, and the project calendar; all of that belongs to the companion paper and is cited, not rebuilt, here (Kilambi Research, 2026).

The rest of the paper proceeds as follows. Section 2 explains how futures, delivery, lease rates, physical premiums, and positioning data actually work, for a non-specialist reader. Section 3 reviews the related literature. Section 4 describes the data and reports summary statistics. Section 5 specifies the macro decomposition, the arbitrage cost build-up, and the positioning metrics. Section 6 reports the results: curves, positioning, delivery, the macro decomposition, and premiums, with explicit falsifiers of the dislocation thesis. Section 7 states what the data cannot show. Section 8 concludes. The data appendix follows the references.

2 Institutional background

2.1 Futures curves, contango, and backwardation

A futures curve relates the price of metal for delivery at different dates. When deferred contracts trade above the front contract, the curve is in contango; when they trade below, it is in backwardation. The standard interpretation, from the theory of storage (Working, 1949; Fama and French, 1987), is that the curve reflects the cost of carrying metal (financing, warehousing, insurance) minus the convenience yield, the implicit value of holding physical metal when stocks are scarce. Deep backwardation signals that holders of physical metal value immediate availability highly, which is the paper market's cleanest tightness signal. Persistent contango signals the opposite: storage is available and no one is bidding aggressively for prompt metal.

The curve's slope can be expressed as an annualized spread: $(\text{deferred}/\text{front} - 1) \times (12/\text{months})$, reported here in basis points. A calendar spread, the price difference between two contract months, is the same object at shorter tenor and is where nearby tightness shows first. Cross-venue spreads (for example COMEX versus LME copper) are not curve slopes; they measure the cost of moving metal between warehouses and regulatory regimes, plus any policy wedge between them.

2.2 Delivery mechanics

Exchange delivery is a warrant transfer, not necessarily a physical movement. On COMEX, metal sits in two categories: registered metal is warranted and committed for delivery against futures; eligible metal meets exchange standards but is not committed and can leave the warehouse system without affecting deliverable supply. A decline in total COMEX stocks that comes entirely from the eligible category does not tighten deliverable supply; a decline in registered stocks does. On the LME, delivery works through warrant cancellation: a warrant holder cancels the warrant to request withdrawal. Cancelled warrants signal intended withdrawal, not completed withdrawal; the metal may remain in the warehouse. Falling on-warrant stocks with a high cancelled-warrant share is the LME analogue of a registered-category draw.

2.3 Lease rates and borrowing costs

A lease rate is the annualized cost of borrowing physical metal for a term, quoted for bullion against the benchmark that publishes them. When nearby physical metal is scarce, lease rates spike: holders of metal demand a premium to lend it. The 1-month lease rate is the standard gauge. Gold forwards (GOFO) were discontinued, and no official body currently publishes current gold or silver lease rates, so lease-rate evidence exists only for documented historical episodes, never as a current reading in this paper.

2.4 Physical premiums

The physical price of metal is the exchange price plus a premium (or minus a discount) that reflects geography, form, and timing. The Yangshan premium is the import premium for copper into China over LME cash; domestic cathode premiums (Shanghai, Guangdong) measure what Chinese buyers pay over the front-month contract for specific grades. Premiums move procyclically with demand

and inversely with price spikes that pause downstream buying; they are a coincident gauge of physical demand, not a fixed tightness gauge.

2.5 Positioning data

The CFTC Disaggregated Commitments of Traders report classifies reportable futures positions into four trader categories: Producer/Merchant/Processor/User (commercial hedgers), Swap Dealers (intermediaries that take the other side of client hedging and investment flows), Managed Money (largely directional speculators), and Other Reportable. Two facts about these categories govern everything said about them in this paper. First, the categories describe the trader's predominant business, not the trade's intent; a swap dealer's gross short is consistent with a fully hedged book, because the offset lives in OTC markets or other venues the report does not see. Second, gross positions and net positions answer different questions: gross shorts show who is present on the short side, net positions show where directional exposure sits, and neither shows intent. This paper never treats gross shorts as manipulation evidence.

The CFTC also publishes a Bank Participation report that would identify bank concentration in gold, silver, and copper positioning. That report was not extracted in this research pass and is a stated gap. The LME publishes no public positioning breakdown equivalent to the COT.

3 Related literature

The theory of storage (Working, 1949) and its futures-pricing formalization (Fama and French, 1987) supply the null hypothesis used throughout this paper: in the absence of scarcity, the curve reflects carrying costs; backwardation and lease-rate spikes are the exceptions that reveal convenience yield, not the rule. That is why Section 6 treats contango as the baseline reading and demands episode-specific evidence before accepting any tightness claim.

The financialization literature (Cheng and Xiong, 2014) documents how index and speculative flows changed commodity futures markets after the mid-2000s, which is the relevant context for reading managed-money positioning alongside prices in Section 6. Erb and Harvey (2006) show that the excess return of commodity futures is largely the roll return, which ties curve shape directly to investor outcomes: persistent contango is a headwind for long futures positions, persistent backwardation a tailwind.

On whether positioning data predicts prices, Sanders and Irwin (2010) test the speculative-bubble claim across commodity futures and find limited evidence that trader positions drive prices, a result consistent with this paper's treatment of COT data as descriptive of who holds risk rather than predictive of where prices go. No paper in this literature, to the desk's knowledge, supplies a validated mapping from CFTC gross shorts to manipulation; that absence is why this paper imposes the interpretive limit stated in Section 2.

4 Data

4.1 Sources and coverage

Prices are IMF Primary Commodity Price System monthly series for the six metals, January 2022 to August 2026, accessed via ElkassabgiData (International Monetary Fund, 2026). Macro series are the Federal Reserve Board H15 10-year nominal Treasury yield and 10-year TIPS constant-maturity yield (month-end, RIFLGFCY10_XII_N.B), the Board's USD nominal effective exchange rate (NEER), and BLS CPI-U (used only for an ex-post real-rate cross-check) (Board of Governors of the Federal Reserve System, 2026; U.S. Bureau of Labor Statistics, 2026). Futures curves are CME official settlements for gold and silver (trade dates September 24–25, 2026) (CME Group, 2026) and TradingView futures-chain snapshots for COMEX copper and LME zinc, nickel, and tin, the last three approximately 147–167 days stale as of September 29, 2026. Positioning is the CFTC Disaggregated COT, futures only, quarterly from 2023-Q1 to 2026-Q3 (180 rows), official annual archives (Commodity Futures Trading Commission, 2026a). Lease rates, premiums, delivery statistics, and episode facts are compiled from exchange data, Reuters and wire reporting, and trade press, with source and evidence grade on every row of the underlying CSVs. Physical-market facts (balances, inventory days of cover) are taken as given inputs from the companion paper's canonical numbers (Kilambi Research, 2026); they are not rebuilt here. Table 1 summarizes the coverage of each series.

Table 1. Data coverage by series

Series	Coverage	Frequency / extent
Metal spot prices (6 metals)	Jan 2022–Aug 2026	Monthly
USD NEER, 10Y TIPS yield	Feb 2022–Jul/Aug 2026	Monthly (55/54 obs)
Gold / silver futures curves	Sep 2026, CME official	Snapshot
Copper futures curve	Sep 2026	Snapshot
Zinc / nickel / tin curves	Apr–May 2026, stale	Snapshot
CFTC Disaggregated COT	2023-Q1–2026-Q3, Au/Ag/Cu	Quarterly, 180 rows
Lease rates	Gold Jan 2025, silver Oct 2025	Event prints only
Physical premiums	Copper current; silver/gold episodic	Spot prints
Delivery statistics	Sep 2026, COMEX and LME	Snapshot
Physical balances	Companion paper	Annual

Notes: Zinc, nickel, and tin rows are explicitly stale or missing; they are reported as such, not used as current evidence.

Source: Desk compilation; see Table 8 for primary sources.

4.2 Summary statistics

Table 2 reports monthly log-return statistics for the six metals and monthly changes in the two macro regressors, February 2022 to August 2026. The sample is short (55 observations) and the period is dominated by a dollar downtrend and a sharp 2025–2026 precious-metals rally; any regression on this window inherits both. The USD NEER row covers February 2022 to July 2026 (54 observations) because the August 2026 NEER print was not in the fetched series vintage.

Silver's 29.9 percent annualized volatility is more than double gold's 13.8 percent, which is the

Table 2. Summary statistics: monthly log returns and macro regressors

Series	n	Mean (%/mo)	Ann. vol. (%)	Min. (%)	Max. (%)
Gold	55	1.62	13.8	-8.01	10.07
Silver	55	1.88	29.9	-14.98	34.29
Copper	55	0.69	16.7	-18.38	9.66
Zinc	55	0.13	22.2	-16.24	14.51
Nickel	55	-0.53	28.4	-18.61	34.54
Tin	55	0.51	26.8	-22.77	17.12
USD NEER % change	54	0.11	4.2	-2.20	2.71
10Y TIPS yield change (pp)	55	0.06	0.93	-0.51	1.01

Notes: Prices are IMF monthly commodity prices, February 2022 to August 2026. Annualized volatility is monthly standard deviation times $\sqrt{12}$. The USD NEER row covers February 2022 to July 2026.
Source: International Monetary Fund (2026), Board of Governors of the Federal Reserve System (2026); desk calculations.

first reason silver's paper market behaves differently from gold's despite their shared monetary role. Nickel's -0.53 percent mean monthly return and 34.54 percent maximum monthly gain sit in the same row as its 2022 squeeze and subsequent structural oversupply; the series contains both stories.

4.3 Curve snapshots

Table 3 gives the curve snapshot used in the analysis. The gold, silver, and copper rows are current as of late September 2026; the zinc, nickel, and tin rows are stale (April–May 2026) and are not treated as current anywhere in this paper.

Table 3. Futures curve snapshots: front versus deferred contract

Metal	Trade date	Front	Deferred	Spread (bps)	Shape
Gold	2026-09-25	Sep 2026, $4,286.2	Aug 2027, $4,471.7	+472	Contango
Silver	2026-09-24	Sep 2026, $63.457	Jul 2027, $65.988	+479	Contango
Copper	2026-09-29	Sep 2026, 6.5660$/lb	Sep 2027, 6.9620$/lb	+603	Contango
Zinc	2026-04-29	Sep 2026, $3,395.8/t	Feb 2027, $3,371.0/t	-175	Backwardation*
Nickel	2026-05-05	Sep 2026, $19,610/t	Mar 2027, $19,789.87/t	+184	Contango*
Tin	2026-04-15	Sep 2026, $50,430/t	Feb 2027, $50,535/t	+50	Flat*

Notes: Spread = (deferred/front − 1) × (12/months between contracts), annualized basis points. Rows marked * are approximately 147–167 days stale as of September 29, 2026; they are not current observations.
Source: CME Group (2026) (gold, silver); TradingView futures chains via desk compilation (copper, zinc, nickel, tin).

4.4 Positioning

Table 4 reports the latest CFTC Disaggregated COT snapshot (September 22, 2026), the basis for Figure 4. The quarterly history (2023-Q1 to 2026-Q3) underlies Figure 3.

Table 4. CFTC Disaggregated COT, September 22, 2026 (contracts)

Metal	Trader category	Gross long	Gross short	Net	Open interest
Gold	Producer/Merchant/Processor/User	17,719	44,496	-26,777	412,800
Gold	Swap Dealer	14,626	250,752	-236,126	412,800
Gold	Managed Money	135,699	8,310	127,389	412,800
Gold	Other Reportable	118,283	19,819	98,464	412,800
Silver	Producer/Merchant/Processor/User	4,047	22,559	-18,512	106,474
Silver	Swap Dealer	22,745	48,987	-26,242	106,474
Silver	Managed Money	19,303	5,994	13,309	106,474
Silver	Other Reportable	15,398	3,263	12,135	106,474
Copper	Producer/Merchant/Processor/User	29,092	141,667	-112,575	301,657
Copper	Swap Dealer	55,201	43,061	12,140	301,657
Copper	Managed Money	96,421	13,899	82,522	301,657
Copper	Other Reportable	32,977	24,977	8,000	301,657

Notes: Futures only. Contract sizes: gold 100 troy oz, silver 5,000 troy oz, copper 25,000 lb.
Source: Commodity Futures Trading Commission (2026a).

5 Methodology

5.1 Macro decomposition

The macro decomposition is a descriptive OLS regression of monthly log returns on two macro factors:

$$r_{m,t} = \alpha_m + \beta_{m,1}\,\Delta\mathrm{USD}_t + \beta_{m,2}\,\Delta\mathrm{TIPS}_t + \varepsilon_{m,t}, \tag{1}$$

where $r_{m,t}$ is the monthly log return of metal m, $\Delta\mathrm{USD}_t$ is the percent change in the Fed Board USD NEER, and $\Delta\mathrm{TIPS}_t$ is the change in the 10-year TIPS constant-maturity yield in percentage points. The sample is February 2022 to August 2026 (n = 55). Reported statistics are the joint R^2, the single-factor R^2 values, and the return correlations with each factor. All regression outputs are (Modelled, C): they are descriptive variance shares on a short sample, not structural estimates. The residual $\varepsilon_{m,t}$ is unexplained variance; this paper does not relabel it as physical scarcity. The physical leg of the verdicts is evidenced separately from inventory, curve, premium, and episode records.

The real rate is the official 10-year TIPS constant-maturity market yield, month-end values ((Observed, A)); an ex-post proxy (nominal yield minus trailing CPI YoY, (Reconstructed, B)) is reported alongside in the data file as a cross-check and is not used in the paper. A data-quality note: the ElkassabgiData catalog freshness stamps were inconsistent with the fetched series (catalogs reporting 2025-06 or 2026-06 while series fetched through 2026-08); catalog stamps were treated as unreliable and the data itself checked.

5.2 Arbitrage-into-delivery cost build-up

A paper-versus-physical spread persists only if the full cost of converting paper into physical exceeds it. The desk's cost build-up for a delivery arbitrage is:

$$\text{Full cost} = \text{spread} - (\text{financing} + \text{freight} + \text{insurance} + \text{conversion} + \text{assay} + \text{warrant} + \text{location} + \text{tariff}), \tag{2}$$

where financing is the carry at the relevant term rate, freight and insurance cover movement between warehouses, conversion covers recasting or refining into deliverable form plus assay, warrant covers exchange warrant and warehouse fees, location covers grade and warehouse differentials, and tariff covers any applicable duty. The build-up is applied qualitatively to the May 2024 copper episode (where the spread exceeded freight and metal flowed) and the early-2025 gold episode (where the binding constraint was conversion, not freight). No complete priced build-up exists for every metal and episode; where a component is unobservable it is stated as unknown rather than assumed.

5.3 Positioning and concentration metrics

Positioning is reported as gross long, gross short, and net by CFTC category, each also expressed as a share of open interest. Concentration is measured by the largest category's gross-short share of open interest and by the managed-money net as a share of open interest. The interpretive rule, stated in Section 2 and enforced throughout, is that gross shorts are composition, not intent: a swap dealer's gross short of 60.7 percent of gold open interest is reported as exactly that, with the offset caveat in the same sentence.

5.4 Retrospective disclosure

This is a retrospective analysis; there was no pre-registered plan. The sample period, the episode list, and the verdict categories were chosen by the desk during the research pass, not fixed in advance. The production note required by house rule appears in the acknowledgments: this paper was produced by Kilambi Research's quantitative research pipeline with AI assistance, and all analysis was verified against source data.

6 Results

6.1 Curve shapes and what they imply

Figure 1 plots the annualized front-to-12-month spread for each metal. Gold (+472 bps) and silver (+479 bps) are in clear contango on official CME settlements from late September 2026 ((Observed, A)). A market pricing acute nearby physical scarcity inverts; neither has. Gold's contango at a record price (IMF August 2026: $4,418.5/oz, (Observed, B)) is the paper market saying the rally is not a scramble for prompt bars. Silver's contango sits alongside the October 2025 lease spike discussed below; the curve normalized after the episode, which is what a liquidity event, as opposed to a structural shortage, looks like in curve space.

Copper is the instructive split. COMEX copper's curve is in steeper contango than either precious metal (+603 bps, (Observed, B)) while the LME cash-to-3-month spread flipped from -$86/t on September 14 to +$26/t backwardation on September 22 and later about +$65/t ((Observed, B)). The September reading is the resolving tail of a genuine squeeze episode: in August 2026, LME

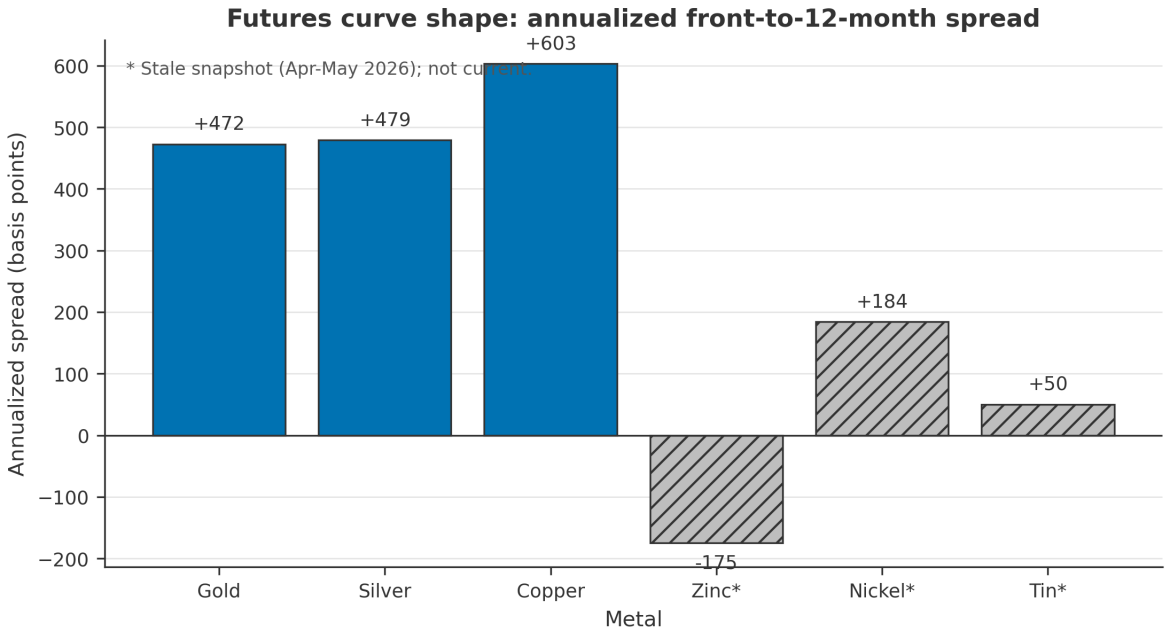


Figure 1. Annualized front-to-12-month curve spread by metal. Starred rows are stale (April–May 2026) and not current.

on-warrant copper fell to about 103 kt on August 19 and cash-to-3-month backwardation exceeded $550/t ((Observed, C)), before stocks rebuilt to about 251 kt by late September ((Observed, C)). The same metal cannot be simultaneously abundant and scarce; the resolution is geography. COMEX sits on record stocks of 696,259 t (69 percent of roughly 1 Mt of exchange-tracked global copper, (Observed, B)), about 60 percent of it in New Orleans warehouses reported near full ((Observed, C)); the LME sits on 255,900 t total with on-warrant stocks of 121,950–133,725 t and 45–51 percent of warrants cancelled for withdrawal ((Observed, B)). The nearby tightness is at the LME prompt and outside the United States; the global exchange-monitored position is the loosest on record. The policy engine is the Section 232 copper investigation: the report was transmitted to the President on June 30, 2025 and was still unpublished as of September 2026, while H1 2026 US refined imports ran at 885,000 t, more than double the H1 2024 pace, and full-year 2025 imports hit a record 1.64 Mt ((Observed, B)). Metal moved to the United States ahead of a tariff that may never arrive; if refined copper is never tariffed, the COMEX stockpile is surplus, not cover. The COMEX-LME spread is as much a tariff bet as a physical signal.

For zinc, nickel, and tin, no current curve statement is possible: the latest located snapshots are 147–167 days stale, and their shapes are reported in Table 3 with that staleness flagged rather than presented as current. Zinc's stale print shows mild backwardation (-175 bps annualized, (Observed, C)), but a five-month-old LME snapshot is not evidence about today's market.

Figure 2 places the five documented 2022–2026 backwardation or dislocation episodes on a single timeline. The pattern across them is that each had a paper-market or policy cause with a physical-market alibi: nickel 2022 was a short-covering and margin event (Section 6.4), copper May 2024 was a commercial-trader short squeeze on thin CME stocks (Section 6.6), the gold and silver lease spikes were transatlantic plumbing and London liquidity events, and the 2026 LME copper prompt is tariff-stockpiling geography. None required, and none left behind, evidence of global physical depletion.

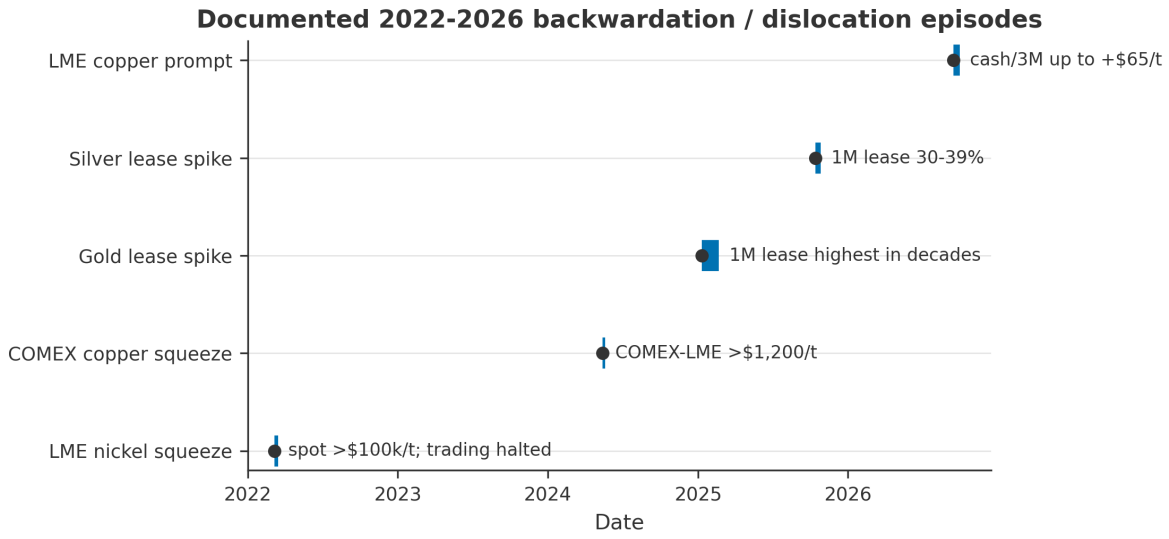


Figure 2. Documented 2022–2026 backwardation and dislocation episodes. Episode windows are approximate; see text for magnitudes.

6.2 Positioning: who is short, and what it does not mean

Figure 3 shows managed-money net positioning (quarterly bars) against spot price (monthly line) for gold, silver, and copper from 2022. Managed money is persistently net long in all three metals through the 2025–2026 rally; there is no positioning evidence of speculative capitulation or of shorts pressing a falling market. Copper managed money did print a net short quarter in early 2024 (about -21,000 contracts) and a small one in early 2025, both resolved back to net long within a quarter, and neither coincided with the May 2024 squeeze, which was driven by commercial shorts, not funds.

Figure 4 decomposes the September 22, 2026 positioning into gross shorts by category (left) and net positions by category (right), each as a share of open interest. The left panel is the one most often misread. Gold swap dealers are gross short 250,752 contracts, 60.7 percent of open interest ((Observed, A)), against gross longs of only 14,626 contracts. That asymmetry is real and it is not evidence of directional bearish exposure or manipulation: swap dealers intermediate, taking the other side of producer hedges and investor longs, and their offsets sit in OTC books and other venues the COT does not see. The CFTC categories do not reveal offsets, so any claim beyond "dealers are net short on the exchange" requires filings or derivatives disclosures the desk does not have. The right panel shows where net exposure actually sits: in gold, managed money net +30.9 percent of open interest and other reportable net +23.9 percent are the long side; the dealer net -57.2 percent is the other side of intermediation, not a bet.

The metal-by-metal composition differs in ways consistent with each market's structure. In silver, producers/merchants are the largest gross-short category (21.2 percent of open interest) with swap dealers close behind (46.0 percent); the producer short is the expected shape of a market where miners and merchants hedge forward production. In copper, producers are the largest gross-short category (47.0 percent of open interest) and swap dealers are net long (+12,140 contracts, (Observed, A)), the reverse of the gold arrangement, which is consistent with dealers warehousing long copper exposure for clients rather than shorting it. None of these patterns is anomalous against the category definitions; the anomalous reading would require evidence the categories do not contain.

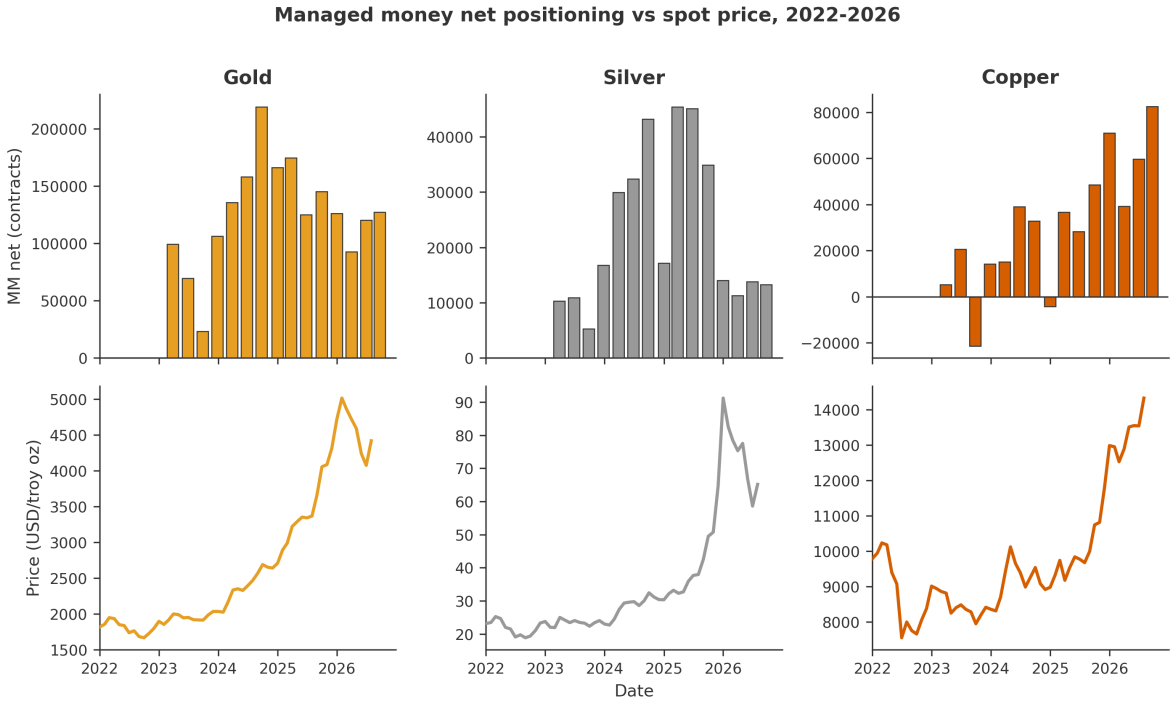


Figure 3. Managed money net positioning (quarterly, bars) versus spot price (monthly, line), 2022–2026. Positioning is CFTC Disaggregated COT, futures only; prices are IMF monthly commodity prices.

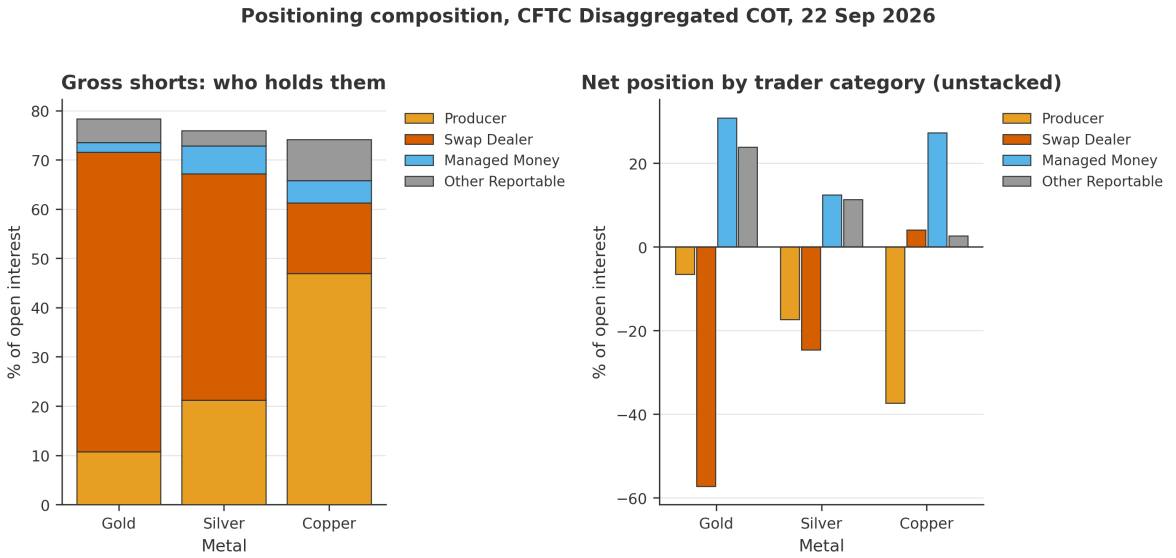


Figure 4. Positioning composition, CFTC Disaggregated COT, September 22, 2026. Left: gross shorts as a share of open interest, stacked by category. Right: net position as a share of open interest by category, unstacked. Gross shorts show who is present on the short side; they do not show intent or offsets.

Concentration risk is a separate question from manipulation, and the honest answer is partial. A single category holding 60.7 percent of open interest gross short in gold is a concentration fact with margin-system implications if that category ever needed to cover quickly; it is not a direction fact. The Bank Participation report, which would identify how much of that dealer short is a handful of banks, was not extracted (stated gap), and the LME has no public COT equivalent, so exchange-level concentration outside COMEX is unmeasured.

6.3 Delivery mechanics and statistics

Figure 5 shows the two delivery pools where September 2026 data exist. The silver panel is the registered-versus-eligible lesson in one chart: COMEX held 330.08 Moz total on September 18, 2026, but only 97.3 Moz of it was registered and deliverable ((Observed, C)); 232.8 Moz was eligible but uncommitted. In the September 10–17 week, 7.1 Moz left the COMEX system, yet registered stocks rose 1.4 Moz because the outflow was eligible metal ((Observed, B)). September delivery notices totaled 6,168, or 30.84 Moz ((Observed, B)), against 97.3 Moz registered: coverage of roughly three times the delivery month's notices. A "vaults are emptying" reading that ignores the registered/eligible split misstates the evidence; deliverable supply rose while total stocks fell. The visible tightening in silver is in exchange-deliverable metal (registered stocks down about 38 percent since October 2025), not in total vaulted metal (LBMA London vaults grew, and the SLV holding of about 492 Moz sits inside the LBMA total and must not be summed with it).

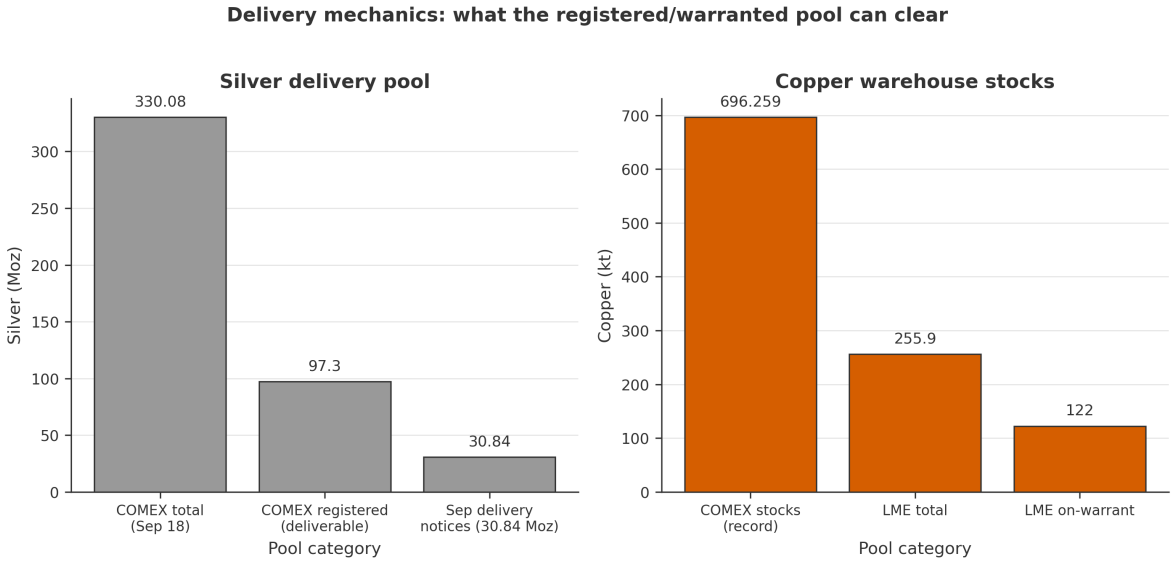


Figure 5. Delivery mechanics, September 2026. Left: COMEX silver; only registered metal is deliverable, and eligible outflows do not tighten deliverable supply. Right: copper warehouse stocks; the LME prompt tightness sits on 122 kt on-warrant against record COMEX stocks.

The copper panel shows the venue split behind Section 6.1: record COMEX stocks against 122 kt on-warrant at the LME. The LME's 45–51 percent cancelled-warrant share means holders were requesting withdrawal of roughly half the warranted metal, but cancelled is intended withdrawal, not completed withdrawal. The binding delivery fact for copper is that the exchange with the tight prompt (LME) and the exchange with the metal (COMEX) are separated by freight, time, and tariff uncertainty, which is exactly what the arbitrage cost build-up in Section 5 predicts will slow

convergence.

Gold exchange inventories close the delivery picture. COMEX held about 852 t on September 3, 2026 ((Observed, C)), while LBMA London vaults held 9,632 t, up 9.1 percent year on year ((Observed, C)). Both are monetary and allocated stocks, not industrial cover: gold clears through price, not quantity, and the January 2025 lease spike was a London-liquidity event, not a stock-out.

6.4 The nickel 2022 squeeze, as history

On Tuesday, March 8, 2022, nickel spiked more than 250 percent intraday to top $100,000/t as Tsingshan Holding Group covered a large short: about 30,000 t held directly on the LME and a further roughly 120,000 t off-exchange in bilateral deals with banks including JPMorgan and Standard Chartered (Reuters, 2026; Bloomberg, 2026; London Metal Exchange, 2026), (Observed, A). The LME halted nickel trading at 8:15 a.m. and retroactively voided all trades executed from midnight to 8:15 a.m., about $3.9 billion of trades by Bloomberg's calculation; Monday's $48,078 close stood. Even at that close, clients of LME brokers had failed to pay about $500 million of margin calls on shorts, Tsingshan accounting for about half. The LME cited systemic risk: four or five clearing members would have failed. Afterward the LME imposed a 15 percent daily price limit and a deferred-delivery fee rule, and Tsingshan's banks agreed a standstill while it covered about 20 percent of the short when prices fell below $30,000/t in mid-March.

Two facts fix the interpretation. The squeeze was a short-covering and margin-call event, not an LME warehouse physical depletion: Tsingshan's bet was that rising Indonesian supply would lower prices, and it was caught by the Russia-Ukraine supply panic. And it struck a deliverable pool that was already small: the LME contract prices Class 1 nickel (at least 99.80 percent purity), which was only about 28 percent of 2024 primary production ((Observed, B)); the 2023–2025 surpluses were mainly Class 2 (ferronickel and NPI) and nickel chemicals, never LME-deliverable. Nickel's price is down about 25 percent since January 2022 (IMF monthly, (Observed, B)), with the 2023–2025 balances in surplus. Invoking 2022 as evidence of current physical tightness gets the direction wrong; it is evidence about margin systems, not metal.

6.5 Macro versus physical: the decomposition

Table 5 reports the decomposition from equation (1) and Figure 6 plots the joint R^2 by metal. Monthly USD and real-rate changes explain 40.5 percent of tin's return variance (the most macro-led, (Modelled, C)), 34.4 percent for copper, 29.7 percent for gold, 19.4 percent for zinc, 18.2 percent for nickel, and 18.0 percent for silver. The USD correlation is negative for all six metals (-0.29 to -0.63, (Modelled, C)); the TIPS-yield-change correlation is negative for gold (-0.27), tin (-0.15), and silver (-0.10) and near zero or slightly positive for copper, zinc, and nickel. In this specification the dollar carries more weight than monthly real-rate changes for every metal.

The verdicts combine the regression with the physical record. Tin is macro-led on the numbers, with the caveat that no physical time series sits behind the regression; the verdict is descriptive, not causal. Gold, silver, copper, and nickel are mixed: macro explains a minority to a third of monthly variance, and each has documented physical-market events (lease spikes, prompt backwardation, the 2022 squeeze) that account for part of the remainder without being relabeled as structural scarcity. Zinc is not identified: the regression explains little, and there is no documented 2022–2026

Table 5. Macro decomposition of monthly log returns, February 2022 to August 2026

Metal	Cum. return (%)	R^2 USD	R^2 TIPS	R^2 joint	Verdict
Tin	32.5	0.392	0.024	0.405	Macro-led
Copper	46.5	0.289	0.000	0.344	Mixed
Gold	143.2	0.294	0.072	0.297	Mixed
Zinc	7.7	0.135	0.005	0.194	Not identified
Nickel	-25.1	0.083	0.029	0.182	Mixed
Silver	181.7	0.174	0.010	0.180	Mixed

Notes: n = 55 months. Regressors: USD NEER percent change and 10-year TIPS yield change. "Mixed" means macro explains part of the variance and documented physical events explain part; "not identified" means neither leg is established. The residual is unexplained, not attributed to physical scarcity.

Source: International Monetary Fund (2026), Board of Governors of the Federal Reserve System (2026); desk calculations.

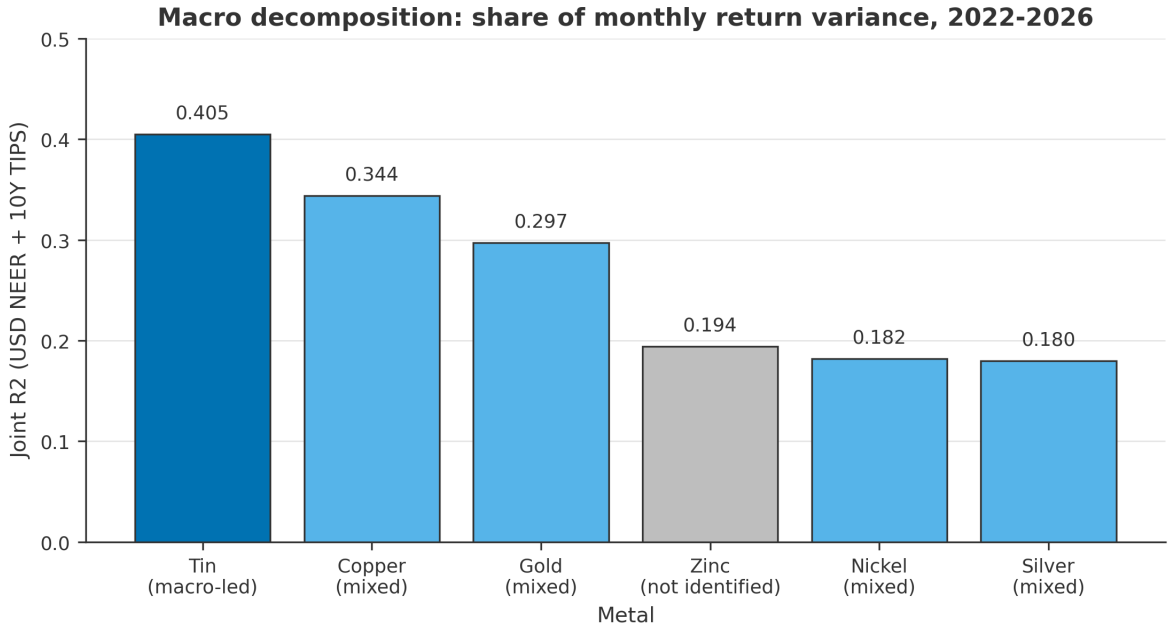


Figure 6. Joint R^2 of monthly returns on USD NEER change and 10-year TIPS yield change, 2022–2026 (n = 55). Tin is the most macro-led; zinc's drivers are not identified by this specification.

dislocation episode for zinc at all, so a paper-tight thesis for zinc has no paper-tight evidence behind it.

The gold result deserves a direct statement because it cuts both ways. Monthly TIPS-yield changes explain only 7.2 percent of gold's return variance (correlation -0.27, (Modelled, C)), even with the official TIPS series. That falsifies the simple "real rates explain gold" version of the macro story at monthly frequency. It does not thereby prove the physical story: gold rose 143.2 percent from January 2022 to August 2026 while lease rates were normal except for the January–February 2025 plumbing episode, and the price level is better explained by the monetary sequence (the Fed cutting to 3.75–4.00 percent in 2025, then hiking back in September 2026; the 10-year real yield near 2.85 percent in late September 2026, (Observed, B)) plus persistent dollar weakness and official-sector demand (central banks bought 863 t in 2025 and an estimated 50 t/month in 2026, (Observed, B)) than by any London stock shortage. The honest statement is that monthly macro changes explain roughly a third of gold's return variance and the rest is unexplained, not automatically physical.

6.6 Physical premiums and the arbitrage tests

Table 6 collects the premium and delivery evidence in one place; the discussion that follows interprets it.

Copper is the only metal with current, sourced physical premiums. The Yangshan import premium printed \$119/t on September 22, 2026, near a four-year high after touching \$124/t (Shanghai Metals Market, 2026), (Observed, B); Shanghai #1 cathode spot sat near 630 CNY/t and Guangdong high-quality near 700 CNY/t on September 19, with Guangdong later falling to 250 CNY/t as copper prices rose and downstream buyers paused ((Observed, B)). The procyclicality is the point: premiums fell when prices rose, because premiums measure coincident physical demand, not a fixed tightness gauge. Codelco's 2026 term offer of \$350/t CIF China (January, (Observed, B)) sits well above January spot import premiums of about \$45/t (S&P Global Commodity Insights, 2026), (Observed, B); term offers embed optionality about future scarcity, and the two can diverge for months.

Silver's Shanghai premium ran about 13 percent over COMEX on September 2, 2026 (Shanghai Metals Market, 2026), (Observed, C) and has not closed via normal arbitrage; China VAT and export licensing are cited as the frictions. The observation is directional, not a priced premium series. Gold wholesale bar premiums are episodic and event-linked: during the January–February 2025 London–New York arbitrage, US 1 kg bars briefly traded +\$40/oz over spot while Bank of England 400 oz bars traded below spot (single-source, (Observed, C)). That inversion is the recasting bottleneck made visible: London 400 oz bars are not COMEX-deliverable and must be recast into 1 kg bars by approved refiners, so the spread priced conversion capacity, not mine supply. Hong Kong kilobars run a low single-digit percent over spot in normal conditions (retail guide, (Observed, C)). Zinc, nickel, and tin current premiums are unknown: Fastmarkets assesses them (Fastmarkets, 2026) but the values are paywalled, and this paper does not infer them from exchange prices.

The full-cost arbitrage test of equation (2) was run by the market itself twice. In May 2024, the COMEX-LME copper premium above \$1,200/t (a record wide against a typical few dollars, (Observed, A)) exceeded freight and logistics costs, so Trafigura and IXM physically shipped copper to the United States and asked producers to divert May–June shipments; changing destination at short notice was "extremely difficult" (Reuters, 2026), the arb worked slowly, and then it overworked: COMEX went from 8,947 t in May 2024 (down 71 percent since late March, the lowest since 2008,

Table 6. Physical premium and delivery evidence, September 2026

Measure	As of	Value	Grade
Copper Yangshan import premium	2026-09-22	$119/t (near 4-yr high)	Observed B
Shanghai #1 cathode premium	2026-09-19	CNY 630/t	Observed B
Guangdong high-quality premium	2026-09-19	CNY 700/t, later CNY 250/t	Observed B
Codelco 2026 term offer, CIF China	Jan 2026	$350/t	Observed B
Silver Shanghai premium over COMEX	2026-09-02	about 13%	Observed C
Gold US 1 kg bar premium	Jan–Feb 2025	+$40/oz over spot	Observed C
COMEX silver registered	2026-09-18	97.3 Moz of 330.08 Moz total	Observed C
COMEX silver delivery notices	Sep 2026	6,168 (30.84 Moz)	Observed B
COMEX copper stocks	2026-09-15	696,259 t (record)	Observed B
LME copper total / on-warrant	2026-09-24/28	255.9 kt / 122–134 kt	Observed B
LME copper cancelled warrants	late Sep 2026	45–51%	Observed B
LME copper cash-to-3M	2026-09-22	+$65/t	Observed B
COMEX gold stocks	2026-09-03	852 t	Observed C
LBMA gold vaults	Aug 2026	9,632 t (+9.1% y/y)	Observed C

Notes: Values are as reported in the prose with evidence grades in the Grade column; see the text for interpretation.

Source: Shanghai Metals Market (2026), S&P Global Commodity Insights (2026), London Metal Exchange (2026), London Bullion Market Association (2026), CME Group (2026), desk compilation.

(Observed, A)) to a record 696,259 t by September 2026. Figure 7 shows both legs: H1 2026 US refined imports of 885,000 t against an implied sub-442,500 t H1 2024 pace, and the resulting COMEX stock record. In early 2025, the gold arb's binding constraint was conversion, not freight: metal moved London to Switzerland and Singapore for recasting before flying to the United States, while US tariff risk (duties rising from 10 to 20 percent on gold and silver refined in some locations, for example Hong Kong) shrank the set of refiners able to service the US. Both episodes confirm the framework's prediction: spreads clear through the cheapest available physical channel, and the channel's frictions, not the spread's size, set the speed.

The silver lease spike of October 2025 (Figure 8) is the strongest nearby-tightness evidence in the sample: 1-month lease rates of 30–39 percent annualized ((Observed, B)) against a normal baseline under 1 percent, London spot trading at a premium to futures, LBMA reporting record-low free float as ETFs encumbered vault stocks, and silver running from above $54 in mid-October to above $80 in December 2025 and a $121.62 peak in January 2026. Lease-rate spikes measure the cost of borrowing nearby metal; they prove nearby tightness at the time, not mine-supply scarcity and not manipulation. The spike cleared: silver shipments from the US and China eased London, and the curve is back in contango. The gold January 2025 episode (1-month lease at its highest in decades, (Observed, B)) reads the same way: tariff-threat London–New York arbitrage plus the recasting bottleneck, not a global mine-supply shortage. Current lease rates for both metals are unknown; LBMA does not publish them, and this paper does not interpolate from the spikes.

6.7 Falsifiers of the dislocation thesis

The thesis that paper markets price something other than physical scarcity is weakened, metal by metal, by the following observable evidence. Each is stated as a condition that would show paper markets pricing physical conditions correctly.

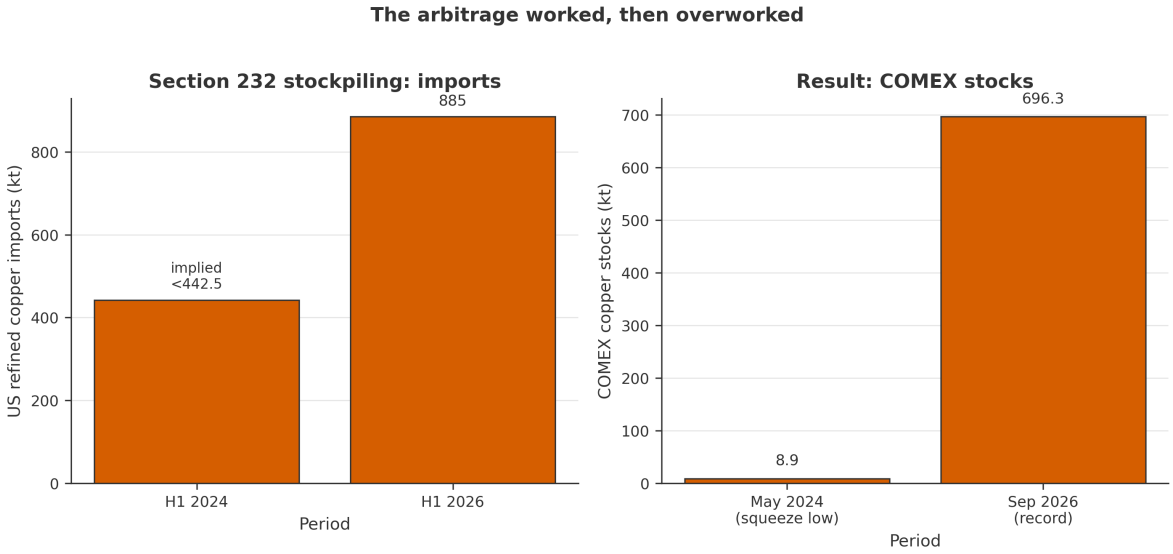


Figure 7. The May 2024 copper arbitrage and its aftermath. Left: US refined copper imports, H1 2024 (implied upper bound from the "more than double" H1 2026 pace) versus H1 2026. Right: COMEX copper stocks at the May 2024 squeeze low versus the September 2026 record.

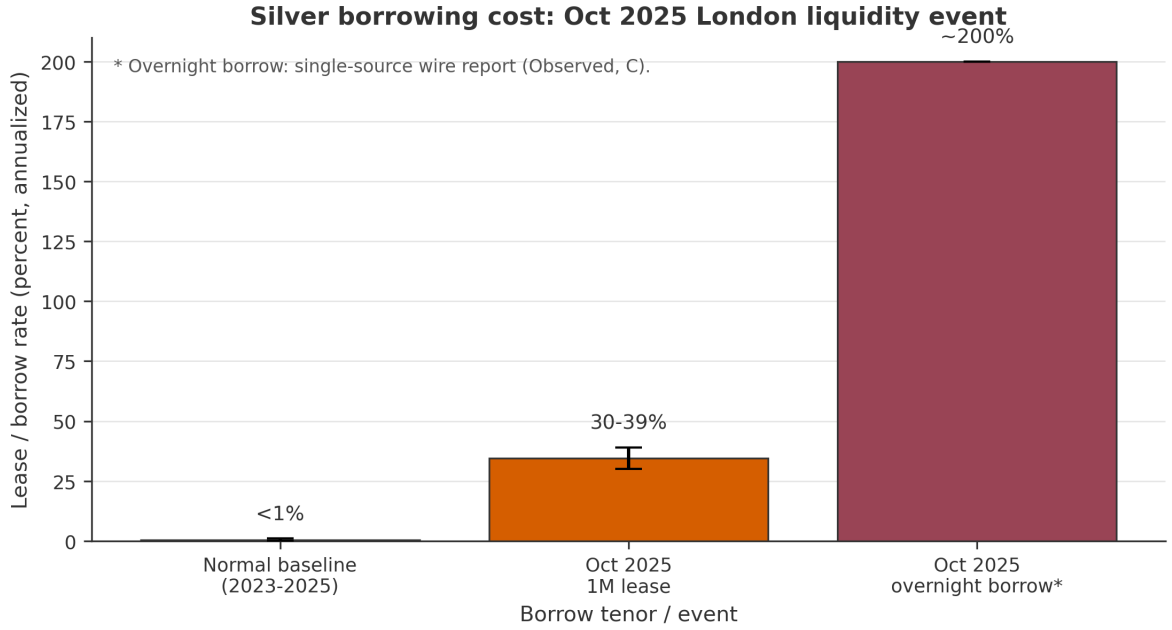


Figure 8. Silver borrowing cost: the October 2025 London liquidity event. The 1-month lease rate printed 30–39 percent annualized against a normal baseline under 1 percent; overnight emergency borrow was reported near 200 percent (single source).

1. Gold and silver curves would invert into sustained backwardation (they are in clear contango at +472/+479 bps annualized on CME, (Observed, A)), which would show the market pricing acute nearby scarcity.

2. Copper's record COMEX stocks would drain without the tariff rationale and LME prompt tightness would spread globally (currently the tightness is regional and policy-driven, not a global shortage).

3. Lease-rate spikes would recur and persist (the gold January 2025 and silver October 2025 episodes were events; current rates are unobservable, and persistence would be the signal, not the episode).

4. The macro decomposition would explain more of return variance (currently USD plus real-rate changes explain 18–41 percent; the residual is unexplained, not automatically physical).

5. Gross dealer and producer shorts would be shown to be unhedged directional exposure (currently the CFTC categories do not reveal offsets; a swap dealer's gross short is not evidence of bearish exposure or manipulation).

6. Zinc would produce a documented dislocation episode (up only about 8 percent since January 2022 with none found; a paper-tight thesis for zinc currently has no paper-tight evidence). LME zinc stocks of 123 kt cover about 3.2 days of demand ((Observed, C)), but ILZSG reported stocks of 739 kt cover about 19.5 days, so the thin exchange read is a warehouse-location fact, not a global one.

7. The May 2024 copper squeeze would have left sustained backwardation (it unwound without it and was followed by a global inventory build; paper squeezes can occur and resolve without physical scarcity).

8. Silver's COMEX outflows would come from registered rather than eligible stocks (the September 2026 outflow was eligible; deliverable supply rose).

7 Limitations and robustness

The sample is short and unusual. Fifty-five months dominated by a dollar downtrend and a 2025–2026 precious-metals rally cannot identify structural relationships; the decomposition is descriptive and its verdicts are evidence-weighted, not causal. The tin "macro-led" verdict rests on the regression alone with no physical time series behind it, and a numeric macro/physical split is not identified for tin or zinc.

Curve evidence is incomplete. Zinc, nickel, and tin have no current curve observation; the stale snapshots are reported, not used. Current gold and silver lease rates are unobservable (GOFO discontinued); the paper reports episodes, not a current reading. Physical premiums for zinc, nickel, and tin are paywalled or unpublished and are not inferred. Gold exchange inventory figures (COMEX 852 t, LBMA vaults 9,632 t) are secondary compilations, (Observed, C); no primary LME or CME series was directly accessed. EFP volumes, which would show how much futures exposure converts to physical off-exchange, were not observable in the data gathered.

Positioning evidence is bounded by what the CFTC publishes. The Disaggregated COT covers futures only, quarterly in the desk's extract; options and OTC offsets are invisible, which is precisely

why gross shorts are not treated as directional evidence. The Bank Participation report was not extracted, so bank-level concentration in gold, silver, and copper positioning is unmeasured. The LME has no public COT equivalent, so non-US positioning concentration is unmeasured.

Table 7 collects the gaps and vintage flags in one place.

Table 7. Data gaps and source vintages

Item	Status
Zinc / nickel / tin curve shapes	Stale snapshots (Apr–May 2026); not current
Current gold and silver lease rates	Unobservable; GOFO discontinued
Zinc / nickel / tin physical premiums	Paywalled (Fastmarkets); not inferred
CFTC Bank Participation report	Not extracted; bank concentration unmeasured
LME positioning	No public COT equivalent
EFP volumes	Not observable in gathered data
USD NEER August 2026	Not in fetched vintage; regression covers Feb 2022–Jul 2026
Primary WSS 2026 table	Secondary summaries only (Observed C)

Notes: Each gap is discussed in the text. Unknowns are not interpolated.
Source: Desk compilation.

The checks run in the adverse direction. The decomposition was checked with an ex-post real-rate proxy (nominal minus trailing CPI YoY); it gives weaker gold/real-rate linkage than the TIPS series, which is why the official TIPS yield is primary, and the choice does not flatter the macro leg. Physical-market inputs (balances, inventory days) are taken from the companion paper's canonical numbers (Kilambi Research, 2026), including its conflict log and stale-data flags; any revision there flows through here. Grade C figures (the copper curve snapshot, the overnight silver borrow print, single-source premiums) are hedged with the number in the prose, not promoted.

8 Conclusion

Paper metals markets in September 2026 are pricing three things, and physical depletion is not the first of them. They are pricing macro risk: the dollar and real rates explain a fifth to two-fifths of monthly return variance, with tin the most macro-led and zinc unidentified. They are pricing regional plumbing: copper's curve is in steep contango globally while the LME prompt touches backwardation, because tariff-stockpiling put the metal in the wrong venue, and the arbitrage that corrects it moves at the speed of freight, conversion capacity, and tariff uncertainty. And they are pricing episodic liquidity stress, in lease-rate spikes and prompt squeezes that have each cleared within weeks to months without leaving sustained backwardation behind.

The per-metal verdicts: gold shows no paper-market dislocation; its price is a macro and official-sector story on a contango curve. Silver shows real but episodic nearby tightness inside a market whose total stocks are ample; the tightening is in deliverable metal, and the lease spike cleared. Copper shows a regional, policy-driven dislocation with a globally loose exchange position. Nickel's 2022 squeeze is margin-system history pointing the wrong way for any current tightness claim. Zinc and tin have no current paper-tight evidence at all, and tin's returns are the most macro-led of the six.

The falsifiers in Section 6.7 are the standing invitation to overturn this. A sustained curve inversion, persistent lease rates, draining COMEX stocks without the tariff rationale, or dealer shorts shown

to be unhedged would each change the verdict. Until that evidence appears, the paper market's answer to the scarcity question is: not scarce here, not scarce now, and where it looks scarce, check the venue.

Acknowledgments

This paper was produced by Kilambi Research's quantitative research pipeline with AI assistance; all analysis was verified against source data. Data work used the shared metals-research workspace; curve, positioning, premium, lease-rate, and episode facts carry source and evidence-grade tags in the underlying CSVs.

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A Data appendix

Table 8 lists every series used, with definition and source. The underlying CSVs (data/ in the metals-research workspace) carry per-row as_of, source, source_url, accessed, and evidence_grade fields per the research protocol.

Table 8. Data appendix: series definitions and sources

Series	Definition	Source
Metal spot prices	IMF Primary Commodity Price System, monthly	International Monetary Fund (2026)
USD NEER	Fed Board nominal effective exchange rate, monthly	Board of Governors of the Federal Reserve System (2026)
10Y nominal yield	Fed Board H15 constant-maturity Treasury yield	Board of Governors of the Federal Reserve System (2026)
10Y TIPS yield	Fed Board H15 TIPS constant-maturity yield, month-end	Board of Governors of the Federal Reserve System (2026)
CPI-U	BLS Consumer Price Index, YoY (ex-post cross-check)	U.S. Bureau of Labor Statistics (2026)
Gold/silver curves	CME official futures settlements, Sep 2026	CME Group (2026)
Copper/zinc/nickel/tin curves	Futures-chain snapshots (stale for Zn/Ni/Sn)	TradingView, via desk
Positioning	CFTC Disaggregated COT, futures only, quarterly	Commodity Futures Trading Commission (2026a)
Bank participation	Not extracted (stated gap)	Commodity Futures Trading Commission (2026b)
LME stocks/notices	Warehouse stock reports and LME notices	London Metal Exchange (2026)
LBMA vaults	London vault holdings (silver/gold context)	London Bullion Market Association (2026)
Lease rates	Episode reports; no current published series	Wire/trade press
Physical premiums	Yangshan, SMM cathode, term offers (copper); Shanghai silver spread; gold bar premiums (episodic)	Trade press (SMM, Platts, Fastmarkets)
Delivery statistics	COMEX stocks, notices; LME warrants/cancellations	CME, trade press
Physical balances	Taken as given from companion paper	Kilambi Research (2026)

Notes: Zinc, nickel, and tin curve snapshots are approximately 147–167 days stale as of September 29, 2026. Current gold and silver lease rates are not publicly observable. Zinc, nickel, and tin physical premiums are paywalled or unpublished.

Source: Desk compilation; primary sources per row in the Source column.